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Comparative Analysis of Machine Learning Models for Palestine Genocide Sentiment

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ABSTRACT — Social media platforms such as X have become major spaces for public discussion of global humanitarian issues, including the Palestinian genocide. The presence of organized information campaigns, commonly known as Hasbara projects, has intensified online discourse and contributed to information warfare through the dissemination of misleading and biased content. The large volume and complexity of such user-generated data make manual analysis impractical, this requiting automated sentiment classification approaches. This study aims to compare the performance of machine learning models in classifying sentiments of tweets related to the Palestinian genocide on media social X. A total 1.702 tweets collected between October 2012 and April 2025 were categorized into three sentiment classes pro, neutral and anti-genocide. Data preprocessing included case folding, noise removal, stopword removal, tokenization, lemmatization, and stemming. Feature extraction was conducted using Term Frequency-Inverse Document Frequency (TF-IDF) and Word2Vec. The evaluated models were Naïve Bayes, Support Vector Machine, Long Short-Term Memory, and Bidirectional Long Short-Term Memory, with hyperparameter fine-tuning applied. The results show that the SVM model achieved the best overall performance, with an accuracy of 85,97% and an f1-score of 85,88%. However, deep learning models, particularly LSTM dan BiLSTM, demonstrated stronger capability in capturing complex contextual sentiment patterns. In conclusion, SVM is the most effective model for this dataset, while deep learning approaches offer promising potential for handling mode complex sentiment representation in future studies.

KEYWORDS — Sentiment Analysis, Machine Learning, Deep Learning, Social Media X, Palestinian Genocide, Text Classification.

I. INTRODUCTION

The Israeli aggression against Palestine has intensified significantly since 7 October 2023, marking one of the most severe escalations in a conflict that has persisted for more than 77 years since 14 May 1948. What initially began as ideological divisions between Israeli settlers and the Palestinian population has evolved into a prolonged occupation that extends beyond a regional dispute, becoming a global geopolitical, humanitarian, and media-driven issue. The events of 7 October 2023, widely referred to as the *Al-Aqsa Flood Operation*, represent a critical turning point that triggered one of the largest escalations of violence in the history of the Israeli occupation of Palestine.

The escalation began with Israeli bombardments targeting civilian settlements in the Gaza Strip, which were subsequently followed by retaliatory actions from Palestinian resistance groups. On the following day, Israel reportedly launched more than 2,000 airstrikes on Palestinian territory within a single day, framing these actions as acts of self-defense. This "self-defense" narrative rapidly spread across international media outlets, where Palestinian fighters were frequently labeled as terrorists, while Israel was portrayed as a defensive actor responding to attacks. Such framing reflects a phenomenon known as the *Hasbara Project*, a digital propaganda strategy employed by Israel to influence and direct global public opinion through information control and media narratives [1].

Since the events of 7 October 2023, public opinion activity on social media platforms has increased dramatically. According to a report by the Arab Center for the Advancement of Social Media, more than 10 million violent-related contents in Hebrew were uploaded across three major social media platforms throughout 2023, with a significant surge after 7 October 2023, reaching an average of 23 posts per minute [2]. In addition, analysis conducted by the Institute for Strategic Dialogue revealed that between 7 October and 24 December 2023, more than 460,000 social media posts contained the keyword "antisemitic" [3]. The term *antisemitic* refers to attitudes, expressions, or actions that demonstrate discrimination against Jewish individuals or communities. These findings highlight the intensity and sensitivity of digital discourse surrounding the Palestinian genocide.

Social media has thus become a primary platform for expressing public opinion on the Palestinian genocide, particularly on Twitter, now known as X, where users tend to be more vocal and explicit in expressing their stances. However, the rapid increase in public discourse on social media is not always accompanied by accurate or verified information. In the context of the Palestinian genocide, the digital space has transformed into a battleground of competing narratives filled with propaganda and disinformation, one of which is driven by the Hasbara Project. This situation makes it increasingly difficult for users to distinguish factual information from narratives shaped by information manipulation.

Therefore, a data-driven approach is required to systematically identify and analyze public attitudes toward the Palestinian genocide. One such approach is sentiment analysis, which enables the classification of public opinion based on textual data. This study aims to classify the sentiment of X users regarding the Palestinian genocide since 7 October 2023 into three categories: pro, neutral, and contra. By applying sentiment analysis using machine learning techniques, this research seeks to provide comparative insights into the performance of different models used in sentiment classification.

Machine learning approaches have been widely applied in previous sentiment analysis studies. Traditional models such as Naive Baye [4] and Support Vector Machine [5] are commonly used due to their simplicity and effectiveness. Meanwhile, deep learning models such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) have demonstrated superior performance in handling large-scale datasets and capturing contextual dependencies in textual data [6]. Accordingly, this study conducts a comparative analysis of machine learning and deep learning models for sentiment classification related to the Palestinian genocide on social media X. The objective of this research is to identify public sentiment toward the Palestinian genocide and to determine the most optimal model for sentiment classification within this humanitarian context.

II. METHODOLOGY

This study adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) as the methodological framework. CRISP-DM is a widely used and structured approach in data mining and machine learning research, consisting of six main phases: *Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation,* and *Deployment*. The framework provides a systematic and iterative process that ensures alignment between research objectives and data-driven analysis. In the context of this study, CRISP-DM is applied to guide the sentiment analysis process of social media X users regarding the Palestinian genocide since 7 October 2023, enabling a structured classification and comparison of machine learning and deep learning models.

A. IDENTIFICATION OF PROBLEMS

The escalation of violence in Palestine since 7 October 2023 has triggered a significant surge in public discourse on social media platforms, particularly on X. Users express diverse and often polarized opinions regarding the Palestinian genocide, which are influenced by propaganda, disinformation, and competing narratives. The rapid growth of user-generated content makes it difficult to manually identify and understand public sentiment accurately and efficiently.

Furthermore, the presence of information manipulation, such as the Hasbara Project, contributes to the distortion of public opinion, complicating the process of distinguishing factual perspectives from biased or misleading narratives. Without a structured analytical approach, identifying dominant public sentiment and understanding its distribution becomes highly challenging. Therefore, the main problem addressed in this study is the lack of an automated, systematic, and data-driven method to classify and analyze public sentiment toward the Palestinian genocide on social media X.

Additionally, previous studies have employed various machine learning and deep learning models for sentiment classification [7]. however, differences in datasets, evaluation metrics, and research contexts make it difficult to determine which model performs optimally in the specific context of the Palestinian genocide discourse. As a result, this study seeks to address the problem of identifying the most effective model for classifying sentiment into pro, neutral, and contra categories within a highly sensitive and narrative-driven social media environment.

B. BUSINESS UNDERSTANDING

This study aims to analyze public sentiment toward the Palestinian genocide as expressed by users on social media X since 7 October 2023. The rapid spread of polarized opinions, propaganda, and disinformation on social media highlights the need for an objective and scalable approach to sentiment classification.

The primary objective of this research is to classify user sentiment into three categories: pro, neutral, and contra toward the Palestinian genocide using sentiment analysis techniques. This approach enables systematic identification of public attitudes and supports comparative evaluation of machine learning and deep learning models.

From a data perspective, social media X provides a large volume of unstructured textual data that reflects real-time public opinion. However, the data contain noise, informal language, and varying expressions, making data understanding a crucial step before preprocessing and modeling.

C. DATA PREPARATION

The Data Preparation phase in the CRISP-DM framework focuses on transforming raw social media data into a structured and suitable format for sentiment analysis. This stage is crucial to ensure data quality and improve model performance. The data preparation process in this study consists of several sequential stages as follows.

1) DATA SCRAPPING STAGES

Data were collected from social media X by retrieving textual content related to the Palestinian genocide since 7 October 2023. The scraping process aimed to gather relevant posts that reflect public opinion and discussions within the defined time frame.

2) LABELING STAGE

The collected data were manually labeled into three sentiment categories: pro, neutral, and contra toward the Palestinian genocide. This labeling process serves as the ground truth for supervised learning and ensures that the sentiment classification reflects the intended research context.

3) DATA BALANCING

To prevent bias during the training process, data balancing was performed to reduce class imbalance among sentiment categories. This step helps ensure that the classification models learn each class proportionally and produce more reliable predictions[8]

4) CASE FOLDING

Case folding was applied by converting all characters in the text data into lowercase. This process aims to standardize the textual data and avoid duplication of terms caused by differences in letter casing [9].

5) CLEANING STAGE

The cleaning stage involved removing irrelevant elements such as URLs, mentions, hashtags, emojis, numbers, punctuation marks, and special characters. This step helps reduce noise and retain only meaningful textual information.

6) TOKENIZATION PROCESS

Tokenization was conducted to split the cleaned text into individual tokens or words. This process allows the text data to be processed and analyzed at the word level.

7) STEMMING

Stemming was applied to reduce words to their root forms. This step helps minimize variations of words with similar meanings and improves the consistency of the feature representation.

8) TRAINING AND TEST DATA SPLITTING PROCESS

The dataset was divided into training and testing subsets. This splitting process allows the models to be trained on one portion of the data and evaluated on unseen data to assess their generalization performance.

9) TERM FREQUENCY-INVERSE DOCUMENT FREQUENCY (TF-IDF)

TF-IDF was used to represent textual data as numerical features for machine learning models. This method measures the importance of a word within a document relative to the entire dataset, enabling effective feature extraction for sentiment classification [10].

10) WORD EMBEDDING

Word embedding techniques were employed to capture semantic relationships between words by representing them as dense vector representations. This approach is particularly useful for deep learning models, as it allows the models to understand contextual similarities within the text data [11].

D. MODELING

The Modeling phase involves selecting and applying appropriate machine learning and deep learning algorithms to classify sentiment in the prepared dataset. In this study, several models were employed to perform sentiment classification in order to compare their performance in the context of the Palestinian genocide discourse on social media X. Traditional machine learning models, namely Naive Bayes and Support Vector Machine (SVM), were used due to their effectiveness and widespread application in text-based sentiment analysis. These models utilize feature representations derived from the TF-IDF weighting scheme [12].

In addition, deep learning models, specifically Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (BiLSTM), were applied to capture contextual and sequential information within textual data [13]. These models leverage word embedding representations to learn semantic relationships between words and improve sentiment classification performance. By employing both machine learning and deep learning approaches, this study aims to conduct a comparative analysis to identify the most optimal model for classifying sentiment into pro, neutral, and contra categories.

E. EVALUATION

Model performance is evaluated using accuracy, precision, recall, and F1-score to measure the effectiveness of sentiment classification into three classes: pro, neutral, and contra. In multiclass classification, these metrics are calculated for each class using a one-versus-rest approach and then averaged to obtain overall performance [14].

Accuracy measures the proportion of correctly classified instances across all sentiment classes, and accuracy can be calculated using (1)

$$Accuracy = \frac{\sum_{i=1}^{3} TP_i}{N} \tag{1}$$

Precision evaluates the correctness of predictions for each class by comparing true positive predictions to all predicted instances of that class, and precision can be calculated using (2)

$$Precision = \frac{TP_i}{TP_i + FP_i} \tag{2}$$

Recall measures the ability of the model to correctly identify all instances belonging to a specific class. And it can be calculated using (3)

$$Recall = \frac{TP_i}{TP_i + FN_i} \tag{3}$$

F1-score is the harmonic mean of precision and recall, providing a balanced evaluation metric. F1-score can be calculated using (4)

$$F1 - score = \frac{2\ x\ Precision_i\ x\ Recall_i}{Precision_i + Recall_i} \tag{4}$$

The final performance of the model is obtained by averaging the precision, recall, and F1-score values across all three sentiment classes, ensuring a comprehensive evaluation of multiclass sentiment classification.

F. DEPLOYMENT

The deployment phase in this study represents the final stage following the completion of the sentiment analysis process. At this stage, the results obtained from the analysis are systematically compiled and documented in the form of a final research report. The report presents the application of sentiment analysis in a practical context and highlights its potential use in real-world scenarios. The outcomes of this study are expected to support data-driven decision-making by providing meaningful insights for users or organizations, thereby demonstrating the practical value of the sentiment analysis approach.

III. RESULTS AND DISCUSSION

A. DATA UNDERSANDING

This study utilized secondary data obtained from social media platform X (formerly Twitter). The data were collected directly from the internet using the Python programming language within the Google Colaboratory environment. The data collection process was conducted through web scraping techniques by retrieving public posts related to the Palestinian genocide issue using predefined keywords and hashtags, such as “Palestine,” “Gaza,” “Genocide,” and other relevant terms.

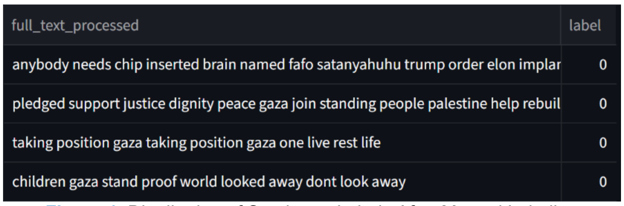
The collected data focused on posts written in English and/or Indonesian and published within the period following October 7, 2023, to capture public reactions after the escalation of violence. In total, 1.700 tweets were successfully collected and used in this study.

At this stage, the dataset consisted of three sentiment categories: pro-genocide, neutral, and contra-genocide. Each tweet was manually labeled to ensure sentiment accuracy. The dataset was then divided into training and testing sets using a 70:30 split, where 70% of the data were used for model training and 30% for model evaluation. This data distribution aimed to ensure balanced learning and reliable performance evaluation of the sentiment classification models.

B. DATA PREPARATION

After the data scraping process, several data preparation steps were conducted to ensure data quality and suitability for sentiment analysis. The collected tweets were first examined to remove duplicate entries, incomplete data, and irrelevant content, resulting in a cleaned dataset ready for further processing.

The labeling stage was then performed manually by categorizing each tweet into three sentiment classes: pro-genocide, neutral, and contra-genocide. This manual labeling process was applied to capture the contextual and semantic nuances of public opinions related to the Palestinian genocide issue, The results of the labeling process are presented in Figure 1.



full_text_processed	label
anybody needs chip inserted brain named fafo satanyahuhu trump order elon implar	0
pledged support justice dignity peace gaza join standing people palestine help rebuil	0
taking position gaza taking position gaza one live rest life	0
children gaza stand proof world looked away dont look away	0

Figure 1. Distribution of Sentiment Labels After Manual Labeling

To address class imbalance within the dataset, a data balancing technique was applied to ensure a more proportional distribution across sentiment categories. This step was intended to prevent model bias toward the dominant class and improve overall classification performance. The sentiment distribution after the balancing process is illustrated in Figure 2.

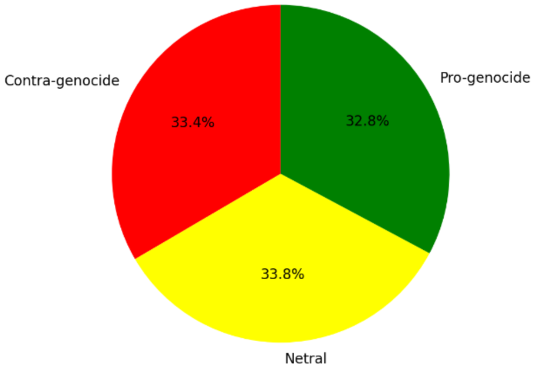


Figure 2. Sentiment Class Distribution After Data Balancing

Text preprocessing was subsequently carried out, starting with case folding, where all characters were converted to lowercase. This was followed by a cleaning process to remove URLs, mentions, hashtags, emojis, punctuation, numbers, and irrelevant symbols. The cleaned text was then processed through tokenization, which segmented sentences into individual tokens to facilitate further linguistic analysis.

Next, stemming was applied to reduce words to their root forms, thereby minimizing word variations and reducing feature dimensionality. This process improved the consistency of textual data for subsequent modeling stages. A comparison of the dataset before and after the preprocessing stages is shown in Table I.

TABLE I
COMPARISON OF TEXT DATA BEFORE AND AFTER PREPROCESSING

Before Preprocessing	After Preprocessing	Sentiment
@sentdefender The whole world stands with Israel and the IDF. End the Islamic regime of Iran.	the whole world stand with israel and the idf end the islamic regim of iran	Pro-genocide
I agree. Israel's war is with Hamas NOT the Palestinian people. Can never understand why the UK won't stand with Palestine. The state of Isreal didn't exist until 1948 when Britain thought it was a good idea to resettle them in Palestine.	i agre israel war be with hama not the palestinian peopl can never understand whi the uk will not stand with palestin the state of israel do not exist until when britain think it be a good idea to resettl them in palestin	Pro-genocide
@EliAfriatISR I stand with Isreal.	i stand with israel	Pro-genocide

After preprocessing, the dataset was divided into training and testing sets using a 70:30 split, where 70% of the data were used for training and 30% for testing. Feature extraction was then performed using Term Frequency–Inverse Document Frequency (TF-IDF) for machine learning models and word embedding techniques for deep learning models to represent textual data numerically.

A. MODELING

This study employed both machine learning and deep learning approaches to perform sentiment classification on social media X data. For the machine learning models, Naive Bayes (NB) and Support Vector Machine (SVM) were implemented using Term Frequency–Inverse Document Frequency (TF-IDF) as the feature representation. TF-IDF was selected to capture the importance of words within the corpus while reducing the influence of commonly occurring terms. To obtain optimal model configurations, GridSearchCV was applied to both NB and SVM models to systematically explore combinations of hyperparameters and select the most suitable settings based on cross-validation results [15].

For the deep learning approach, Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) models were utilized due to their capability to capture sequential dependencies and contextual information in textual data. These models employed Word2Vec word embedding to represent semantic relationships between words in a dense vector space. The embedding layer was fine-tuned during training to adapt to the specific characteristics of the dataset. Model parameters, including embedding dimensions, number of hidden units, dropout rates, batch size, and training epochs, were determined through a manual tuning process to achieve stable convergence and effective learning.

B. EVALUATION

The evaluation stage aims to assess the performance of each sentiment classification model using standard multi-class evaluation metrics. In this study, model performance was evaluated using accuracy, precision, recall, and F1-score, which were computed based on the confusion matrix for three sentiment classes: pro-genocide, neutral, and contra-genocide. To ensure robust and unbiased evaluation, 10-fold cross-validation was applied to all models. Table II presents the comparative evaluation results of all models used in this study.

TABLE I
PERFORMANCE COMPARISON OF SENTIMENT CLASSIFICATION MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)

Naïve Bayes	78.90	79.99	78.87	79.09
SVM	85.97	86.14	85.88	85.86
LSTM	80.09	80.11	80.08	79.71
BiLSTM	77.90	77.84	77.84	77.95

The Naive Bayes model achieved an average accuracy of 78.90% with a balanced performance across precision, recall, and F1-score, indicating stable classification results despite its simplicity. The Support Vector Machine (SVM) model demonstrated the best overall performance, achieving an average accuracy of 85.97% and an F1-score of 85.86%, outperforming all other models. This result indicates that SVM is highly effective in capturing sentiment patterns in high-dimensional TF-IDF feature space.

For deep learning approaches, the LSTM model achieved an average accuracy of 80.09% and an F1-score of 79.71%, showing competitive performance compared to traditional machine learning models. Meanwhile, the BiLSTM model obtained an average accuracy of 77.90% with an F1-score of 77.95%. Although BiLSTM is theoretically capable of capturing bidirectional contextual information, its performance was slightly lower, likely due to the limited size and variability of the dataset.

Further analysis using confusion matrices revealed that all models classified the neutral sentiment class with the highest accuracy, while most misclassifications occurred between the pro-genocide and contra-genocide classes, indicating semantic similarity between opposing opinions. Additionally, ROC curve and AUC analysis showed that all models achieved AUC values above the baseline for each sentiment class, confirming their capability to perform multi-class sentiment classification effectively.

Overall, the evaluation results indicate that the SVM model provides the most optimal performance in this study, surpassing both traditional and deep learning-based models. This finding suggests that, for sentiment classification tasks with structured textual features and moderate dataset sizes, machine learning models such as SVM can outperform more complex deep learning architectures.

C. DEPLOYMENT

The deployment phase represents the final stage of the CRISP-DM framework, where the outcomes of the sentiment analysis process are presented and documented. In this study, deployment does not involve the development of a real-time system but focuses on the interpretation and utilization of the analysis results. The findings obtained from the sentiment classification models are compiled into a structured research report that summarizes public sentiment patterns regarding the Palestinian genocide issue on social media X.

The results of this study can be utilized as a reference for researchers, policymakers, and organizations to better understand public opinion dynamics in digital spaces. By identifying sentiment tendencies and the most optimal classification model, this research provides a data-driven foundation for further studies and decision-making related to social media analysis and public discourse monitoring.

IV. CONCLUSION

This study conducted a comparative analysis of machine learning and deep learning models for sentiment classification related to the Palestinian genocide issue on social media X. The sentiment analysis task classified public opinions into three categories: pro-genocide, neutral, and contra-genocide. Several models were evaluated, including Naive Bayes, Support Vector Machine (SVM), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM), using standardized preprocessing techniques and evaluation metrics.

The experimental results demonstrate that the SVM model achieved the best overall performance, with the highest accuracy and F1-score compared to the other models. This finding indicates that SVM is more effective in handling high-dimensional textual features generated by TF-IDF in datasets with moderate size. Although deep learning models such as LSTM and BiLSTM are capable of capturing sequential and contextual information, their performance was slightly lower, likely due to dataset characteristics and limited data variability.

Overall, this study highlights that traditional machine learning approaches can outperform more complex deep learning models in certain sentiment classification scenarios. Future research may explore larger and more diverse datasets, alternative word embedding techniques, or hybrid models to further improve sentiment classification performance.

CONFLICTS OF INTEREST

The authors have no personal interests related to the issues or topics discussed in this manuscript. The data collection procedures, methodology, and interpretation of the results were conducted without influencing any system. In other words, the validity and independence of the results presented are proven

AUTHORS' CONTRIBUTIONS

Conceptualization, Wafa Tsabita and Afrida Helen ;methodology, Wafa Tsabita; software, Wafa Tsabita; validation, Wafa Tsabita, Afrida Helen and Mira Suryani; formal analysis, First Author's Name; investigation, Wafa Tsabita; resources, Wafa Tsabita; data curation, Wafa Tsabita; writing—original draft preparation, Wafa Tsabita; writing—reviewing and editing, Wafa Tsabita; visualization, Wafa Tsabita; supervision, Wafa Tsabita; project administration, Wafa Tsabita; funding acquisition, Afrida Helen.

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